1. Data Flow Diagrams (DFDs)

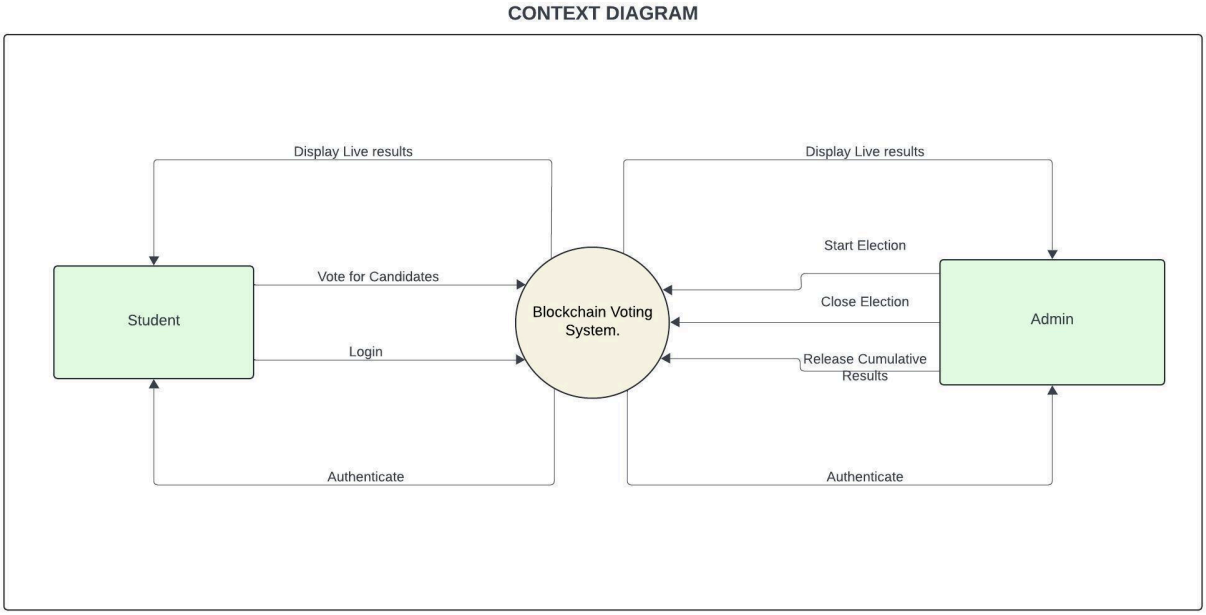

Fig. 1: Context Diagram (Level 0 DFD)

The context diagram in Flg. 1 visually represents the interactions between the key actors and the core system in the university's election process.

The diagram highlights two primary actors: Students and Admin. *Students* interact with the system to **Vote for Candidates** and **Display Live Results**, ensuring their participation and access to real-time information. The *Admin*, on the other hand, has the authority to **Start Election**, **Close Election**, and **Release Cumulative Results**, managing the overall election lifecycle. The use of blockchain technology within the system aims to enhance security, transparency and trust in the election process.

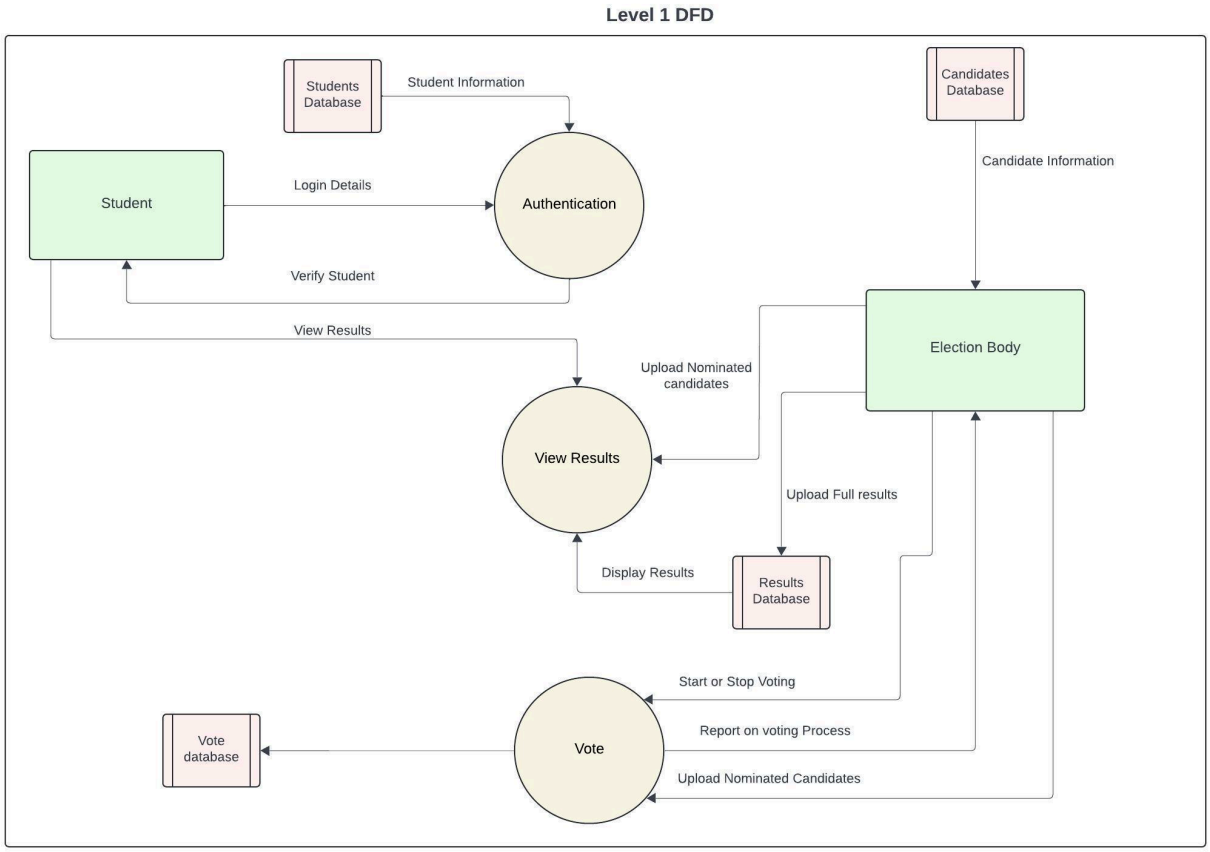

Fig. 2: Level 1 Data Flow Diagram (DFD)

The Level 1 DFD in Fig. 2 above provides a more detailed view of the system's processes compared to the context diagram. It breaks down the system into several key processes:

1. **Authentication:** This process verifies the identity of both Students and the Election Body by checking their Login Details against stored information.
2. **Verify Student:** This process ensures that only registered students are allowed to participate in the voting process.
3. **Upload Nominated Candidates:** This process allows the Election Body to add the list of candidates running for election.
4. **Vote:** This process enables registered students to cast their votes for their preferred candidates. The votes are stored in the Vote Database.
5. **View Results:** This process allows both Students and the Election Body to view the current election results.
6. **Upload Full Results:** Once the voting period ends, the Election Body uses this process to upload the final and official election results.
7. **Start** or **Stop Voting:** This process gives the Election Body control over the voting period, allowing them to start or stop voting as needed.

8. **Report on Voting Process:** This process enables the Election Body to generate reports on various aspects of the voting process, such as voter turnout and election statistics.

Key Data Flows

The DFD also illustrates the flow of data between different components:

- **a.** Student Information and Candidate Information are used for authentication and verification purposes.
- **b.** Login Details are used to authenticate users.
- **c.** Nominated Candidates are uploaded by the Election Body.
- **d.** Votes are cast by students and stored in the Vote Database.
- **e.** Results are displayed to both Students and the Election Body.
- **f.** Reports on the voting process are generated by the Election Body.

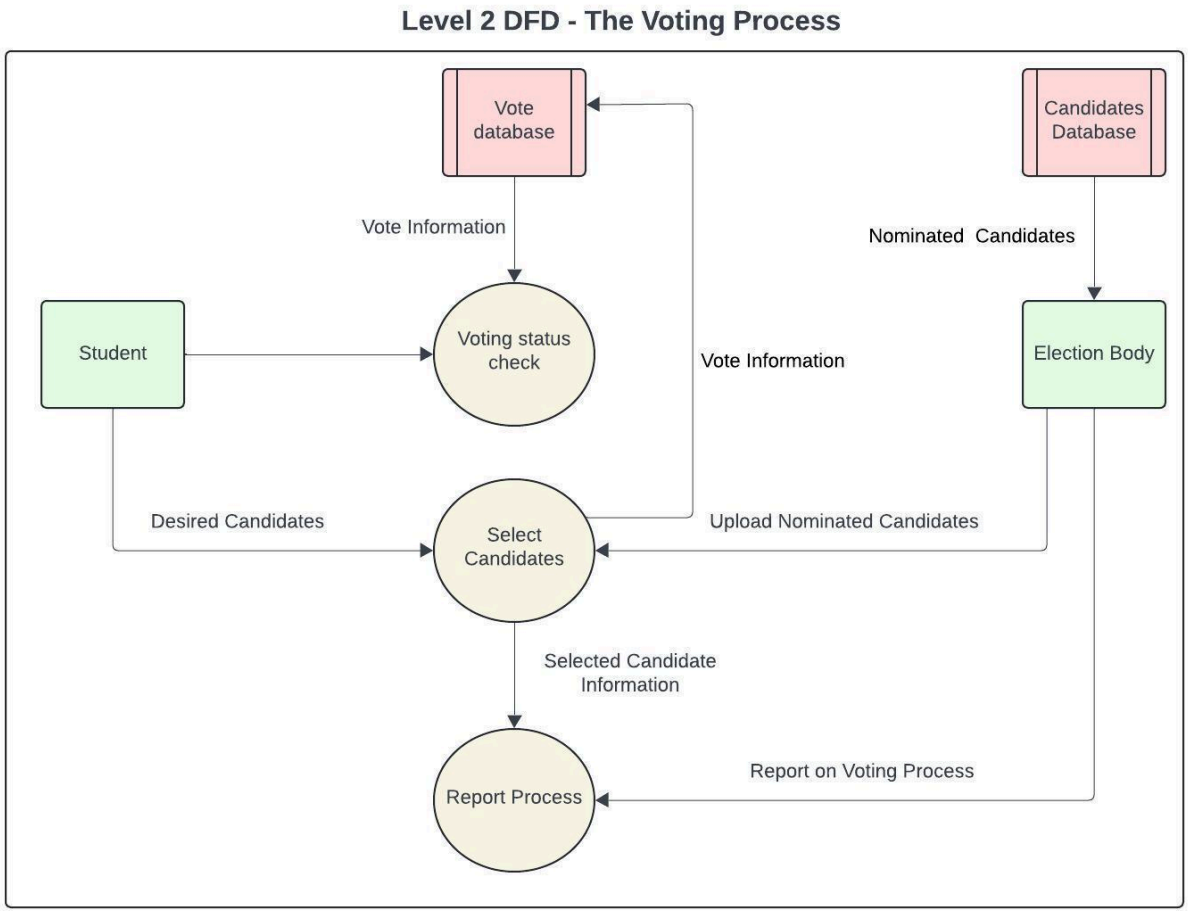

Fig. 3: Level 2 Data Flow Diagram (DFD) - Voting Process

The Level 2 DFD in Fig. 3 zooms into the **Vote** process from the Level 1 DFD, providing a more detailed view of the steps involved in casting a vote.

1. **Voting Status Check:** This process verifies whether the student is eligible to vote by checking their voting status.
2. **Select Candidates:** This process allows the student to select their desired candidates from the list of nominated candidates.
3. **Vote Information:** Once the student has selected their desired candidates, their vote information, including their student ID and the selected candidates, is recorded in the Vote Database.
4. **Report Process:** This process generates reports on various aspects of the voting process, such as the number of votes cast and the number of votes received by each candidate.

Key Data Flows

a. Nominated Candidates are retrieved from the Candidates Database.
b. Student information and voting status are used to verify eligibility.
c. Selected Candidate Information is recorded in the Vote Database.
d. Reports on the voting process are generated based on the recorded vote information.

2. Sequence Diagram

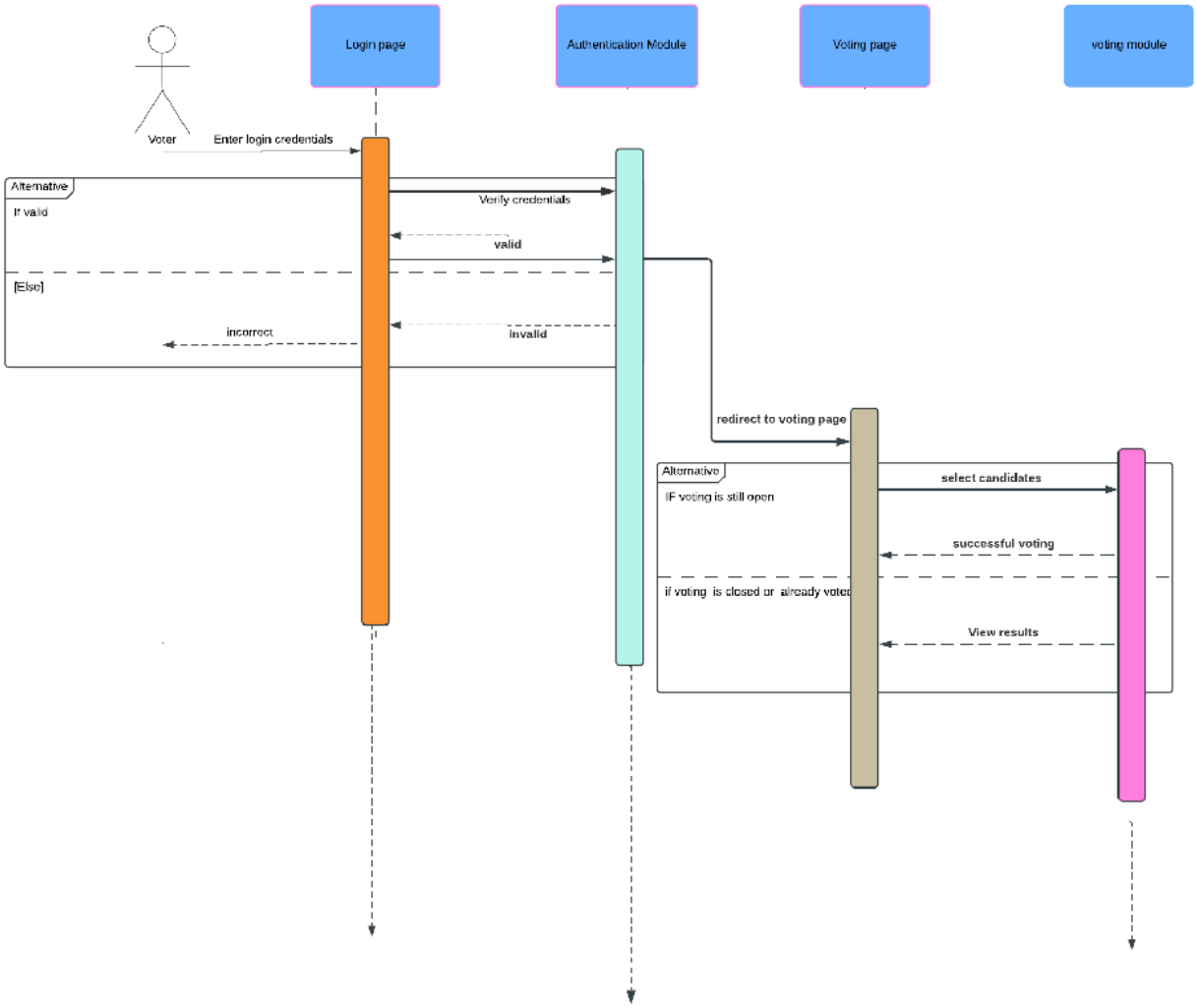

Fig. 4: Sequence Diagram

This Sequence Diagram in Fig. 4 illustrates the interaction between different components in the voting system when a voter attempts to cast their vote.

1. **Voter Interaction:** The sequence begins with the Voter entering their login credentials on the Login page.
2. **Authentication:** The Login page sends the entered credentials to the Authentication Module for verification.
3. **Credential Validation:** The Authentication Module checks the validity of the credentials.

4. **Authentication Result:**
    a. If the credentials are valid, the Authentication Module sends a **valid** message to the Login page.
    b. If the credentials are invalid, the Authentication Module sends an ***invalid*** message to the Login page, and the process ends with an ***incorrect*** message to the Voter.
5. **Voting Page Access:** If the credentials are valid, the Login page redirects the Voter to the Voting page.
6. **Voting Status Check:** The Voting page checks the voting status.
7. **Voting Period:**
    a. If voting is still open, the Voter is allowed to proceed to select candidates.
    b. If voting is closed or the Voter has already voted, the Voting page redirects the Voter to the ***View results*** page.
8. **Candidate Selection:** The Voter selects their desired candidates on the Voting page.
9. **Successful Voting:** The Voting page records the Voter's vote and displays a ***successful voting*** message.

3. Entity Relationship Diagram (ERD)

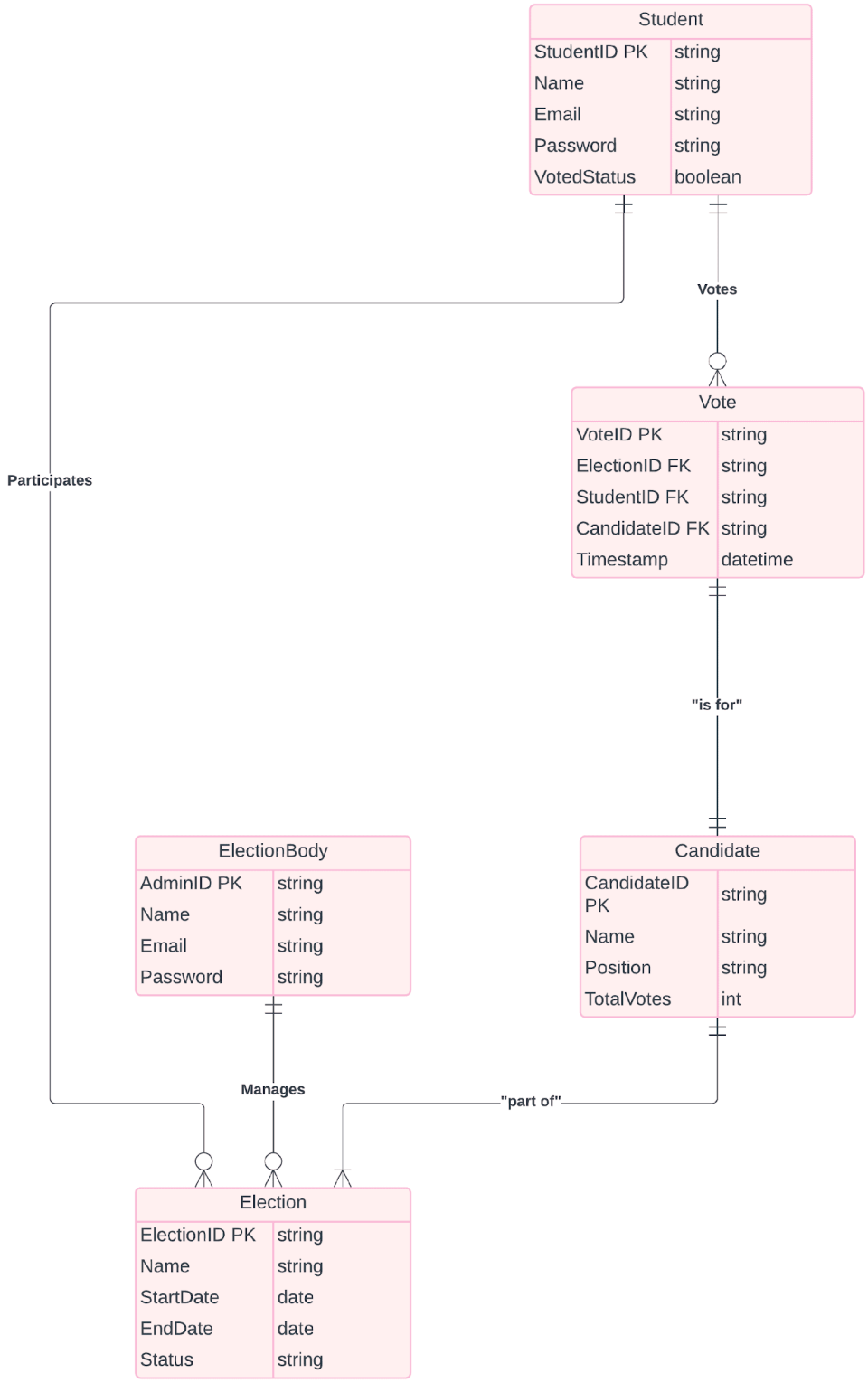

Fig. 5: Entity Relationship Diagram (ERD)

This ERD in Fig. 5 above represents the database schema for the university's secure election system. It captures the entities involved and their relationships.

Entities:

1. **Student:** Represents a student in the university. Attributes include StudentID *(primary key)*, Name, Email, Password and VotedStatus *(boolean to indicate if the student has already voted)*.
2. **ElectionBody:** Represents the administrative body responsible for managing elections. Attributes include AdminID *(primary key)*, Name, Email and Password.
3. **Candidate:** Represents a candidate running in an election. Attributes include CandidateID *(primary key)*, Name, Position and TotalVotes.
4. **Election:** Represents an election event. Attributes include ElectionID *(primary key)*, Name, StartDate, EndDate and Status.
5. **Vote:** Represents a single vote cast by a student. Attributes include VoteID *(primary key)*, ElectionID *(foreign key referencing Election)*, StudentID (*foreign key referencing Student)*, CandidateID *(foreign key referencing Candidate)*, and Timestamp.

Relationships:

- **Participates:** A many-to-many relationship between Student and Election, indicating that a student can participate in multiple elections, and an election can have multiple participating students. This relationship is implemented through the Vote entity.
- **Manages:** A one-to-many relationship between ElectionBody and Election, indicating that an ElectionBody can manage multiple elections.
- **"is for":** A many-to-many relationship between Candidate and Election, indicating that a candidate can run in multiple elections, and an election can have multiple candidates. This relationship is also implemented through the Vote entity.

Key Points:

- The use of foreign keys in the Vote entity establishes the relationships between Student, Candidate and Election.
- The VotedStatus attribute in the Student entity ensures that a student cannot vote multiple times in the same election.
- The Timestamp attribute in the Vote entity records the time at which a vote was cast.